

AI and Biotechnology/Bioinformatics

Assignment 2:

QSAR Data Curation using ChEMBL

Course: AI and Drug Discovery (2026)

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QSAR Data Curation Summary

1. Introduction

The objective of this assignment was to perform bioactivity data retrieval, evaluation, and curation from the ChEMBL database for use in Quantitative Structure–Activity Relationship (QSAR) modeling. QSAR modeling relies on high-quality, standardized bioactivity data associated with a well-defined molecular target. This work focused on assessing target suitability, retrieving relevant bioactivity records, and applying essential preprocessing steps required for QSAR analysis.

The molecular target selected for this assignment was **BRCA1 (Breast Cancer Type 1 Susceptibility Protein)**. BRCA1 plays a crucial role in DNA damage repair and is of significant clinical relevance in breast and ovarian cancers, making it an important target in cancer biology research.

2. Target and Data Selection

Selected Target

- **Target Name:** BRCA1 (Breast Cancer Type 1 Susceptibility Protein)

Bioactivity data associated with BRCA1 were explored using the ChEMBL database to evaluate their suitability for QSAR modeling. The focus was placed on standardized bioactivity measurements, particularly IC₅₀ values, which are commonly used in QSAR studies.

Data Evaluation and Justification

- **Number of available bioactivity records:** 5 IC₅₀ measurements

The analysis revealed that BRCA1 has a very limited number of directly associated small-molecule bioactivity records in ChEMBL. Although this dataset size is insufficient for building a

statistically robust QSAR model, it was retained for this assignment to demonstrate the complete QSAR data retrieval and curation workflow for educational purposes. The limited data availability highlights the challenges associated with QSAR modeling for targets that are not traditionally druggable.

3. Data Curation Workflow

Bioactivity data were retrieved programmatically from the ChEMBL database using Python. The following preprocessing steps were applied to ensure data consistency and quality:

- Retrieval of BRCA1-associated bioactivity records using the ChEMBL API
- Selection of standard bioactivity measurements (IC50)
- Removal of records with missing, invalid, or non-numeric values
- Standardization of bioactivity units to nanomolar (nM)
- Elimination of duplicate or inconsistent entries
- Export of the curated dataset into CSV format for downstream analysis

4. Workflow Overview

Step-by-step workflow:

1. Selection of BRCA1 as the molecular target
2. Retrieval of bioactivity data from the ChEMBL database
3. Evaluation of data availability and quality
4. Filtering and preprocessing of IC50 bioactivity records
5. Standardization and final dataset preparation

Conceptual Workflow Diagram:

Target Selection → ChEMBL Data Retrieval → Data Evaluation → Data Cleaning & Standardization → Curated Dataset

5. Conclusion

This assignment successfully demonstrates the QSAR data curation workflow using **BRCA1** as a case study. Although the limited number of available bioactivity records restricts robust **QSAR model development**, the exercise highlights the importance of target selection, data availability assessment, and careful preprocessing in QSAR studies. The curated dataset serves as a proof-of-concept for QSAR data preparation methodologies.

6. GitHub Repository

GitHub Repository Link:

https://github.com/Abeeraiftikhar/AI_and_Drug_Discovery_Course_2026.git